Computation for Analytics

DIANE WOODBRIDGE, PH.D

Course Delivery

Flipped Classroom

- A type of active learning pedagogy.
- Assign recorded lectures so that students can learn on their own pace and utilizes the classroom time for more exercises and discussion.
- Make sure to watch and submit a review quiz (your quiz review grades don't matter, but submissions will count toward your grades).
- Post questions that you are not clear about on Piazza.
- Also post topics that you want me to re-cover during the class on Piazza.

About Computation for Analytics (MSDS501)

Class Overview

Day 1
Class Intro, Basic Computer Architecture,
Terminal and Shell Commands

Day 2
Version Control & Git, Programming & Python,
Data Science Packages/Libraries, Debugging, Error Handling

Day 3
Python Data Types, Conditional Statements, Loop, pytest

Day 4
Python File I/O, Data Aliasing, Function

Day 5
Programming Patterns

Day 6
Packages, Libraries and Modules

Day 7
numpy

Day 8
Object Oriented Programming

Day 9
PEP8 and PEP20, Problem Solving

Day 10
Programming in Cloud Computing

About Computation for Analytics (MSDS501)

Discussion Session Schedule

Days	Topics
July 7, 10	Day 1, Day 2
July 14, 17	Day 3, Day 4
July 21, 24	Day 5, Day 6
July 28, 31	Day 7, Day 8
August 4, 7	Day 9, Day 10

About Computation for Analytics (MSDS501)

Evaluation Criteria

- Attendance and Professionalism - 10 %
- Weekly Assignments - 20 % (Due: 9pm on Thursdays)
- Quiz - 70 % (1-2pm on Fridays : Must be IN-PERSON, Remotely taken quiz will receive 0.)

Grader : Maneel Reddy (mkarri@dons.usfca.edu)

Weekly Assignment

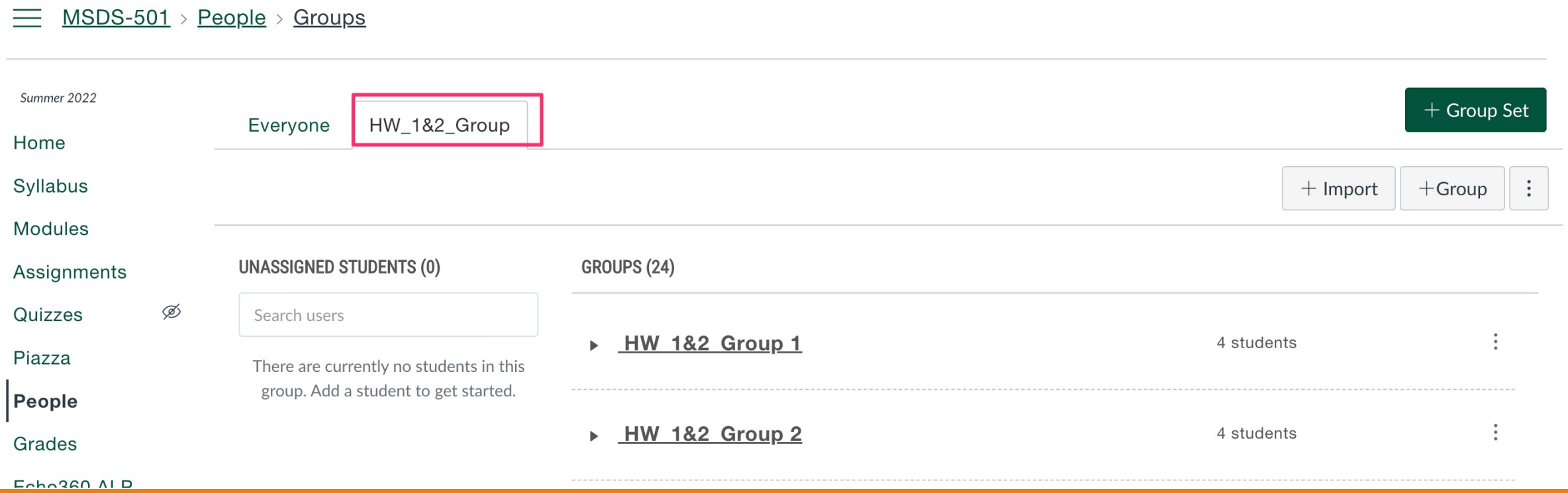
All assignment for this course will be “Group Assignment” and you will have a group for HW_1_2 and HW_3_4_5.

- I will assign up to 3 ppl for group and you will submit one assignment per group. (Not every member needs to submit)
  - You can consider it as your study group.
  - For building networks, I will randomly assign groups.
- If you do not work as a group, please let me know today.
  - However, I found that working individually didn’t help my student who has a computer science degree - especially for building networks in the program.
- I am going to ask participation rate and your grade will be weighed based on it.
  - Tip : Make sure to do your homework as much as possible prior to your group meeting.

Weekly Assignment

I created on Canvas - Do not create your own group (it is not visible to me)

- If you want to work individually, please let me know by today.



About Computation for Analytics (MSDS501)

This class is a standard, graded course with letter grades A - F.

- I consider an A grade to be above and beyond what most students have achieved.
- A B grade is an average grade for a student or what you could call "competence" in a business setting.
- A C grade means that you either did not or could not put forth the effort to achieve competence.
- Below C (F) implies you did very little work or had great difficulty with the class compared to other students.

The expected final score for this course is 85 ± 3 (and close to normal distributions).
The following grades will be given if the class grade distribution falls within the expectation.
However, if the grade distribution does not meet the aforementioned criteria, grades will be curved.

Score	Letter Grades
90 - 100	A+, A and A-
80 - 90	B+, B and B-
70 - 80	C+, C and C-
Below 70	F

For the students who are experts in programming



I have not yet seen students who are masters in programming.

- This is an introductory course to make everyone on the same page, and I expect you to master all the materials. Please review the materials thoroughly, as doing well in the boot camp courses are essential for proceeding to the program and being placed in your dream practicum.
- However, if you need more advanced materials, please get a Premium Leetcode account and start practicing.
  - Start with the Learn section and move forward.

Questions/comments about the courses?

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Announcement

Get to know each other - Please make sure to sign up by August 10th

Assignment 1 (Group HW) - Due by Thursday, July 13th 9 PM

- If you want to work individually for bootcamp, please let me know by today.
- One file submission by any group member is fine.
- You can submit multiple times by the deadline.

Quiz 1 - Friday, July 14th 1pm - 2pm

- Study list and practice quiz are available on Canvas.
- Include multiple choices and programming (Both closed-book and Honorlock is required).

Questions about the assignments?

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Day 1 - Content

Basic Computer Architecture

Files and Folders

Terminal, Shell Commands/Scripts

Homebrew/Anaconda/Python Installation

Questions

Chmod permission string - user, group, others

Why added VSCode path to ~/.zsh_profile?

- https://code.visualstudio.com/docs/setup/mac

When poll is active, respond at **pollev.com/msds**

Text **MSDS** to **37607** once to join

From which one does the CPU retrieve data directly?

- Disk
- Memory
- Thread
- AWS
- None of the above

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

When poll is active, respond at **pollev.com/msds**

Text **MSDS** to **37607** once to join

From which one does the CPU retrieve data directly?

- Disk
- Memory
- Thread
- AWS
- None of the above

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Poll locked. Responses not accepted.

From which one does the CPU retrieve data directly?

- Disk
- Memory
- Thread
- AWS
- None of the above

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

When poll is active, respond at **pollev.com/msds**

Text **MSDS** to **37607** once to join

When $pwd returns /opt/homebrew/anaconda3 what is the output of $pwd after $cd include/c++/v1

- /opt/include/c++/v1
- /include/c++/v1
- /opt/homebrew/anaconda3/include/c++/v1
- include/c++/v1
- None of the above

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

When poll is active, respond at **pollev.com/msds**

Text **MSDS** to **37607** once to join

When $pwd returns /opt/homebrew/anaconda3 what is the output of $pwd after $cd include/c++/v1

/opt/include/c++/v1

/include/c++/v1

/opt/homebrew/anaconda3/include/c++/v1

include/c++/v1

None of the above

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Poll locked. Responses not accepted.

When $pwd returns /opt/homebrew/anaconda3 what is the output of $pwd after $cd include/c++/v1

/opt/include/c++/v1

/include/c++/v1

/opt/homebrew/anaconda3/include/c++/v1

include/c++/v1

None of the above

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

When poll is active, respond at **pollev.com/msds**

Text **MSDS** to **37607** once to join

When $pwd returns /opt/homebrew/anaconda3, what is the output of $pwd after $cd ../../anaconda3/envs?

- /anaconda3/envs
- /opt/anaconda3/envs
- /opt/homebrew/anaconda3/envs
- None of the above

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

When poll is active, respond at **pollev.com/msds**

Text **MSDS** to **37607** once to join

When $pwd returns /opt/homebrew/anaconda3, what is the output of $pwd after $cd ../../anaconda3/envs?

/anaconda3/envs

/opt/anaconda3/envs

/opt/homebrew/anaconda3/envs

None of the above

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Poll locked. Responses not accepted.

When $pwd returns /opt/homebrew/anaconda3, what is the output of $pwd after $cd ../../anaconda3/envs?

- /anaconda3/envs
- /opt/anaconda3/envs
- /opt/homebrew/anaconda3/envs
- None of the above

Start the presentation to see live content. For screen share software, share the entire screen. Get help at pollev.com/app

When poll is active, respond at **pollev.com/msds**

Text **MSDS** to **37607** once to join

For the given code, what is the output?

```
nums=(10 11 12 13)
threshold=12

for num in ${nums[*]}
    do
    if [ $num -le $threshold ]
    then
        echo $num
    else
        echo "?"
    fi
done
```

```
10
11
12
?
```

```
10
11
12
```

13

None of the above

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

When poll is active, respond at **pollev.com/msds**

Text **MSDS** to **37607** once to join

For the given code, what is the output?

```
nums=(10 11 12 13)
threshold=12

for num in ${nums[*]}
    do
    if [ $num -le $threshold ]
    then
        echo $num
    else
        echo "?"
    fi
done
```

- 10 11 12 ?
- 10 11 12
- 13
- None of the above

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Poll locked. Responses not accepted.

For the given code, what is the output?

```
nums=(10 11 12 13)
threshold=12

for num in ${nums[*]}
    do
    if [ $num -le $threshold ]
    then
        echo $num
    else
        echo "?"
    fi
done
```

- 10
11
12
?
- 10
11
12
- 13
- None of the above

Start the presentation to see live content. For screen share software, share the entire screen. Get help at pollev.com/app

Shell Scripting

Command Substitute

- You can use backtick(`) or $() to substitute shell commands in your shell script.
  - Ex. `$(command)`, `` `command` ``

https://pubs.opengroup.org/onlinepubs/9699919799/utilities/V3_chap02.html#tag_18_06_03

Example 03

ex03_input.sh includes lines of shell command.

Read this file and print the output of each command.

Day1 topics that you want me to go over again?

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Day 2 - Contents

Version Control

- Git

Programming

Python

- Python Development
- Python Libraries

Debugging

Error/Exception Handling

When poll is active, respond at **pollev.com/msds**

Text **MSDS** to **37607** once to join

Choose a command that updates the existing local git repository to up-to-date?

- git init
- git push
- git commit
- git pull
- None of the above

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

When poll is active, respond at **pollev.com/msds**

Text **MSDS** to **37607** once to join

Choose a command that updates the existing local git repository to up-to-date?

- git init
- git push
- git commit
- git pull
- None of the above

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Poll locked. Responses not accepted.

Choose a command that updates the existing local git repository to up-to-date?

- git init
- git push
- git commit
- git pull
- None of the above

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

When poll is active, respond at **pollev.com/msds**

Text **MSDS** to **37607** once to join

You can execute a .ipynb file by $python file_name.ipynb

True

False

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

When poll is active, respond at **pollev.com/msds**

Text **MSDS** to **37607** once to join

You can execute a .ipynb file by $python file_name.ipynb

True

False

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Poll locked. Responses not accepted.

You can execute a .ipynb file by $python file_name.ipynb

True

False

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

When poll is active, respond at **pollev.com/msds**

Text **MSDS** to **37607** once to join

For the following code what is the output?

```
try:
    a = 1
    b = "something"
    if(a < b):
        print("b is bigger")
    else:
        print("a is bigger")
except TypeError:
    print("Inappropriate data type")
except ArithmeticError:
    print("Wrong arithmetic operation")
except NameError:
    print("Name not found")
```

- b is bigger
- a is bigger
- Inappropriate data type
- Wrong arithmetic operation
- Name not found

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

When poll is active, respond at **pollev.com/msds**

Text **MSDS** to **37607** once to join

For the following code what is the output?

```
try:
    a = 1
    b = "something"
    if(a < b):
        print("b is bigger")
    else:
        print("a is bigger")
except TypeError:
    print("Inappropriate data type")
except ArithmeticError:
    print("Wrong arithmetic operation")
except NameError:
    print("Name not found")
```

b is bigger

a is bigger

Inappropriate data type

Wrong arithmetic operation

Name not found

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Poll locked. Responses not accepted.

For the following code what is the output?

```
try:
    a = 1
    b = "something"
    if(a < b):
        print("b is bigger")
    else:
        print("a is bigger")
except TypeError:
    print("Inappropriate data type")
except ArithmeticError:
    print("Wrong arithmetic operation")
except NameError:
    print("Name not found")
```

b is bigger

a is bigger

Inappropriate data type

Wrong arithmetic operation

Name not found

Start the presentation to see live content. For screen share software, share the entire screen. Get help at pollev.com/app

Debugging

Let’s debug count_min_max.py

Day 2 topics that you want me to go over again?

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**